

Day 3 — Mon 7 Sep

Countdown: 19 days to General English & GK

Method: start by recalling yesterday's topic with this page closed. Study the sections below, work every example by hand, then do 20 MCQs.

● General — précis

Source: General/01_GENERAL_PAPER_PLAYBOOK.md

3. PRÉCIS WRITING — 10 marks

3.1 What a précis is, in plain English

You are given a passage of maybe 350–450 words. You must rewrite it in about **one-third** of its length, keeping *every essential idea* and *nothing else*, in **your own words**, in **your own continuous prose**, and give it a **title**.

It is not a summary you write casually. It has strict conventions, and most of the 10 marks are for **following the conventions**, not for elegance. This is the single most *mechanically learnable* item on the English half of the paper. A candidate who memorises the rules should score 7–8/10 reliably.

3.2 The strict rules

#	Rule	Note
1	Length $\approx$ one-third of the original.	If the passage is 360 words, write 115–125. Tolerance is roughly $\pm 10\%$. Being 50% too long is the commonest failure.
2	Always give a title.	Short — 3 to 7 words. Written above, centred or left, underlined. A missing title costs a mark outright.
3	Write in the third person.	"I believe..." in the passage becomes "The writer believes..." or simply "It is argued that...". Never carry "I", "we", "you" across.
4	Write in the past tense — for narrative and reported opinion.	Universal truths and permanent facts stay in the present ("Water boils at 100°C"; "The Himalayas lie to the north").
5	Use your own words.	Some technical terms must be reused (you cannot paraphrase "photosynthesis"). But do not lift whole sentences.
6	No new material.	No examples of your own, no opinions of your own, no facts from outside the passage.
7	No direct speech, no quotations, no dialogue.	Convert to indirect speech.
8	One continuous paragraph.	A précis is never bulleted and rarely more than one paragraph.
9	Keep the author's proportion and order.	If a third of the passage is about causes, roughly a third of your précis is about causes.
10	No rhetorical questions, no exclamations.	Convert "Is this acceptable?" to "The writer questions whether this is acceptable."

11	Cut all: examples, illustrations, repetitions, figures of speech, digressions, adjectives of emphasis.	This is where your compression comes from.
12	State the word count at the end in brackets.	e.g. (118 words). Shows you obeyed rule 1.

3.3 Step-by-step method (the 25-minute procedure)

Step 1 — Read once, fast (2 min). Do not write. Just get the gist. Ask: *what is this passage arguing?*

Step 2 — Read again with a pencil (4 min).

- Underline the **topic sentence** of each paragraph.
- **Strike through** every example, every "for instance", every list of illustrations, every quotation, every repetition.
- Circle **connective words** (however, therefore, moreover) — they carry the logical skeleton and you must preserve the logic, if not the words.

Step 3 — Count the original (1 min).

Count the words in one full line, multiply by the number of lines. Divide by three.

Write your target on the rough sheet: e.g. **Target: 120 words (±12).**

Step 4 — Write the skeleton in the margin (3 min).

Reduce each paragraph to one clause. Four paragraphs → four clauses. That is your précis in miniature.

Step 5 — Draft in rough (6 min). Join the clauses into flowing prose. Third person, past tense. Do not look back at the passage while drafting — this is what forces "own words".

Step 6 — Count and trim (3 min). Over the target? Cut adjectives first, then merge sentences with participles ("Having considered X, the writer argues Y").

Step 7 — Write the fair copy + title (5 min). Title last, once you know what the passage is really about. Add (N words).

Step 8 — Run the checklist (1 min). See §3.6.

3.4 Compression techniques (how you actually get to one-third)

Technique	Before	After
Phrase → word	"in spite of the fact that"	"although"
List → category	"rice, maize, millet and wheat"	"cereals"
Clause → participle	"He finished the report and then submitted it."	"Having finished the report, he submitted it."
Drop the illustration	"Pollution is severe — in Delhi, for example, ..."	"Pollution is severe."
Two sentences → one	"The scheme failed. It lacked funds."	"The scheme failed for want of funds."
Drop intensifiers	"extremely important and highly significant"	"important"
Nominalise	"when the government decided to intervene"	"on government intervention"

3.5 Worked example

The passage (352 words)

Every year, in the hill districts of the North-East, the rains arrive with a ferocity that outsiders find difficult to imagine. Roads that took three years to build vanish in a single night. Culverts collapse. Villages that were a two-hour

drive from the district headquarters become, for weeks at a time, effectively islands. I have watched a schoolteacher in one such village walk eleven kilometres along a ridge because the road below had simply ceased to exist.

The usual response is to blame the contractor. Sometimes that blame is deserved. But the deeper problem is that we have been building the wrong kind of road in the wrong kind of place. Engineering practice in India was largely codified for the plains, where the chief enemies are traffic load and temperature. In young, folded, heavily-rained-upon mountains, the chief enemy is water — water moving through soil, water undercutting slopes, water finding every seam in the rock. A road that is merely strong is not enough. A road must also drain.

There is, moreover, a question of who builds. For decades, hill roads have been designed in distant offices by engineers who have never walked the alignment in August. Local knowledge — which slope slips, which spring runs all year, where the old footpath ran and why it ran there — is treated as folklore rather than as data. Yet villagers can often predict a landslide site with an accuracy that would embarrass a soil report.

The remedy is not more money alone, though more money is needed. It is a change of method: design that begins with drainage rather than ending with it; specifications written for mountains rather than borrowed from the plains; and a genuine role for the village in choosing and maintaining its own alignment. Where communities have been given charge of small assets — as some states in the region have experimented with — maintenance has often improved dramatically, because the asset belongs to somebody. Until that shift happens, we will keep rebuilding the same road, and the rain will keep taking it away.

Model précis (117 words — one-third of 352 ≈ 117)

Title: Why Hill Roads Fail

The writer observed that heavy monsoon rain in North-Eastern hill districts repeatedly destroyed newly built roads and cut villages off for weeks. While contractors were often blamed, he argued that the deeper fault lay in method: Indian engineering practice was framed for the plains, where load and temperature matter most, whereas in young, high-rainfall mountains the decisive threat is water. A road therefore had to drain as well as bear weight. He further noted that alignments were designed remotely by engineers unfamiliar with the terrain, and that villagers' knowledge of unstable slopes was dismissed. The remedy he proposed was drainage-led design, mountain-specific specifications, and community responsibility for choosing and maintaining roads.

(117 words)

What that précis did, move by move

- **Title** given, short, captures the argument (not just the subject).
- "I have watched a schoolteacher..." — a pure illustration — **deleted entirely**.
- First person "I", "we" → **"the writer"**, third person throughout.
- Past tense for the writer's reported argument ("argued", "noted", "proposed"); but the general truth "the decisive threat is water" kept in the present. Both are correct.
- Order of the original (problem → misdiagnosis → real cause → who builds → remedy) **preserved exactly**.
- No new material: the "some states in the region" clause was generalised, not turned into a named example.
- Rhetorical richness ("the rain will keep taking it away") dropped — précis is plain.

3.6 The pre-submission checklist

Tick all nine before you move on:

- [] Title written and underlined
- [] Length within $\pm 10\%$ of one-third
- [] Word count stated in brackets
- [] Third person — no "I", "we", "you", "our"
- [] Past tense for reported argument; present only for permanent truths
- [] One continuous paragraph, no bullets, no headings
- [] No quotations, no direct speech, no rhetorical questions
- [] No examples, no illustrations, no repetition
- [] Nothing in it that was not in the passage

3.7 Practice plan

Find any 300–450 word passage — a newspaper editorial is ideal — and précis it in 25 minutes. **Do six of these across the 21 days.** Editorials from *The Hindu*, *Indian Express* or *Nagaland Post* work perfectly and double as current-affairs revision. Precision comes from repetition, not from reading about précis.

● Core — Boolean algebra

Source: `_Shared/CORE_01_DigitalLogic.md`

C1. Boolean algebra

Concept

Ordinary algebra deals with numbers that can take infinitely many values. **Boolean algebra deals with variables that can take exactly two values, 0 and 1**, and three operations: AND ($\cdot$), OR ($+$), NOT ($'$). It was invented by George Boole in 1854 as a formalisation of logic; **Claude Shannon** realised in 1938 that switching circuits obey exactly the same algebra, and that insight is the foundation of all digital design.

The whole practical point is: *a circuit is an expression, an expression is a circuit*. If you can simplify the expression algebraically you have literally made the circuit smaller and cheaper. That is why "minimisation" is a syllabus item in its own right.

Postulates and laws — the complete table

Every law below has a **dual**, obtained by swapping AND $\leftrightarrow$ OR and $0 \leftrightarrow 1$. This is the **Principle of Duality** and it halves what you have to memorise.

#	Law	AND form	OR form (dual)
1	Identity	$A \cdot 1 = A$	$A + 0 = A$
2	Null / Dominance	$A \cdot 0 = 0$	$A + 1 = 1$
3	Idempotent	$A \cdot A = A$	$A + A = A$
4	Complement	$A \cdot A' = 0$	$A + A' = 1$
5	Involution	$(A')' = A$	(same)
6	Commutative	$A \cdot B = B \cdot A$	$A + B = B + A$

7	Associative	$(AB)C = A(BC)$	$(A+B)+C = A+(B+C)$
8	Distributive	$A(B+C) = AB + AC$	$A + BC = (A+B)(A+C)$ ← no decimal analogue!
9	Absorption	$A(A+B) = A$	$A + AB = A$
10	Absorption (2nd form)	$A(A'+B) = AB$	$A + A'B = A + B$
11	De Morgan	$(AB)' = A' + B'$	$(A+B)' = A' \cdot B'$
12	Consensus	$AB + A'C + BC = AB + A'C$	$(A+B)(A'+C)(B+C) = (A+B)(A'+C)$

Law 8's OR-form and Law 10 are the two that students forget and that exams exploit.

$A + A'B = A + B$ — verify: if $A=1$ both sides are 1; if $A=0$, LHS = $0 + 1 \cdot B = B$, RHS = $0+B = B$ ✓

De Morgan's theorems — the most important result in the topic

Statement (2 variables):

$(A \cdot B)' = A' + B'$ "NOT of AND = OR of NOTs"
 $(A + B)' = A' \cdot B'$ "NOT of OR = AND of NOTs"

Statement (n variables / generalised):

$(A_1 \cdot A_2 \cdot \dots \cdot A_n)' = A_1' + A_2' + \dots + A_n'$
 $(A_1 + A_2 + \dots + A_n)' = A_1' \cdot A_2' \cdot \dots \cdot A_n'$

The plain-English rule you should actually use: "Break the bar, change the sign."

To complement an entire expression: complement every variable **and** swap every AND with an OR and every OR with an AND (and swap 0s and 1s).

Proof by truth table (write this out if asked to "prove"):

A	B	AB	(AB)'	A'	B'	A'+B'	A+B	(A+B)'	A'B'
0	0	0	1	1	1	1	0	1	1
0	1	0	1	1	0	1	1	0	0
1	0	0	1	0	1	1	1	0	0
1	1	1	0	0	0	0	1	0	0

Columns (AB)' and A'+B' are identical; columns (A+B)' and A'B' are identical. **Hence proved.**

Why it matters practically: De Morgan is what lets you convert *any* circuit into a NAND-only or NOR-only circuit — which is what real ICs are built from because NAND is the cheapest gate in CMOS.

Worked example 20 — algebraic simplification

Simplify $F = A'B'C + A'BC + AB'C + ABC' + ABC$

$F = A'B'C + A'BC + AB'C + ABC' + ABC$
 $= A'C(B' + B) + AB'C + AB(C' + C)$ [factor 1st two; factor last two]
 $= A'C(1) + AB'C + AB(1)$ [complement law]
 $= A'C + AB'C + AB$
 $= C(A' + AB') + AB$ [factor C]
 $= C(A' + B') + AB$ [law 10: $A' + AB' = A' + B'$]
 $= A'C + B'C + AB$

Check by counting minterms: original covers m1,m3,m5,m6,m7. $A'C = m1,m3$; $B'C = m1,m5$;

$AB = m_6, m_7$. Union = $\{1, 3, 5, 6, 7\}$ ✓

Worked example 21 — applying De Morgan

Find the complement of $F = A(B + C'D)$

$$\begin{aligned} F' &= [A(B + C'D)]' \\ &= A' + (B + C'D)' && \text{[De Morgan on the AND]} \\ &= A' + B' \cdot (C'D)' && \text{[De Morgan on the OR]} \\ &= A' + B' \cdot (C + D') && \text{[De Morgan on the inner AND, and involution } C''=C\text{]} \\ &= A' + B'C + B'D' \end{aligned}$$

Duality vs complement — a distinction examiners test

	Dual	Complement
Operation	Swap AND↔OR, 0↔1	Swap AND↔OR, 0↔1, AND complement every variable
Example on $F = A + B'C$	$F^d = A(B' + C)$	$F' = A'(B + C')$
Relationship	$F^d(A, B, C) = F'(A', B', C')$	—

C2. Logic gates

The seven gates

Gate	Symbol expr	Truth (A,B → Y) 00,01,10,11	Plain-English rule	Hand-drawn shape
AND	$Y = A \cdot B$	0,0,0,1	Output 1 only if all inputs 1	Flat back, D-shaped front
OR	$Y = A + B$	0,1,1,1	Output 1 if any input 1	Curved back, pointed front
NOT	$Y = A'$	(1 in → 0 out)	Inverter	Triangle + bubble
NAND	$Y = (A \cdot B)'$	1,1,1,0	AND then invert	AND shape + bubble
NOR	$Y = (A + B)'$	1,0,0,0	OR then invert	OR shape + bubble
XOR	$Y = A \oplus B = A'B + AB'$	0,1,1,0	Output 1 if inputs differ / odd number of 1s	OR shape + extra curved line
XNOR	$Y = (A \oplus B)' = AB + A'B'$	1,0,0,1	Output 1 if inputs same — an equality/comparator	XOR + bubble

XOR properties — heavily examined

$A \oplus 0 = A$ $A \oplus 1 = A'$ (XOR with 1 = controllable inverter)
 $A \oplus A = 0$ $A \oplus A' = 1$
 $A \oplus B = B \oplus A$ $(A \oplus B) \oplus C = A \oplus (B \oplus C)$ (commutative, associative)
 $A \oplus B \oplus C = 1$ when an ODD number of inputs are 1 → parity generator
If $A \oplus B = C$ then $A \oplus C = B$ and $B \oplus C = A$ → basis of simple encryption/RAID parity

Universal gates

Concept: NAND and NOR are called **universal** because *any* Boolean function can be built from NAND alone, or from NOR alone. That is commercially decisive: a chip fab only has to perfect one gate. In CMOS, NAND is preferred (its series-nMOS/parallel-pMOS structure is faster than NOR's for the same area).

NAND realisations (memorise the counts — MCQs ask "how many NAND gates?"):

Function	Using NAND	Count
NOT A	NAND(A, A)	1
A·B	NAND then invert: NAND(NAND(A,B), NAND(A,B))	2
A+B	NAND(A', B') = NAND(NAND(A,A), NAND(B,B))	3
$A \oplus B$	—	4
$A \odot B$ (XNOR)	—	5

NOR realisations:

Function	Count
NOT	1
A+B	2
A·B	3
XNOR	4
XOR	5

Note the pleasing symmetry: NAND is cheap for AND, NOR is cheap for OR; XOR costs 4 NANDs but 5 NORs, and XNOR is the reverse.

Why they are universal: because {AND, OR, NOT} is a functionally complete set, and NAND can produce all three (rows 1–3 above). Same argument for NOR.

Converting any SOP circuit to all-NAND: replace every AND gate by a NAND and every OR gate by a "bubbled-OR" — which by De Morgan *is* a NAND. So a two-level AND-OR circuit maps one-for-one to a two-level NAND-NAND circuit with the same gate count. **Similarly, any two-level OR-AND (POS) circuit maps directly to NOR-NOR.**

IC / logic family facts (syllabus mentions "Integrated Circuits")

Scale	Gates per chip	Examples
SSI (Small Scale Integration)	1–10	7400 quad NAND, 7404 hex inverter
MSI (Medium)	10–100	Decoders, MUX, counters, registers (74138, 74151, 7490)
LSI (Large)	100–10,000	Small memories, early processors
VLSI (Very Large)	>10,000 (now billions)	CPUs, GPUs, SoCs
ULSI	>1,000,000	Modern microprocessors

Family	Full name	Speed	Power	Noise immunity
TTL	Transistor–Transistor Logic (74xx)	Fast	High	Moderate

CMOS	Complementary MOS (40xx, 74HCxx)	Moderate-fast	Very low static	High
ECL	Emitter Coupled Logic	Fastest	Highest	Low

- **Fan-in** = number of inputs a gate can accept. **Fan-out** = number of gate inputs one output can drive (typical TTL fan-out = 10).
- **Propagation delay** t_{pd} = time from input change to output change. Total delay of a circuit = delay along the **longest (critical) path**.
- **Noise margin** = the voltage cushion between a valid output level and the input threshold. CMOS has the best.

Common IC numbers worth knowing

IC	Function
7400	Quad 2-input NAND
7402	Quad NOR
7404	Hex inverter
7408	Quad AND
7432	Quad OR
7486	Quad XOR
7447	BCD to 7-segment decoder/driver
7474	Dual D flip-flop
7476	Dual JK flip-flop
7483	4-bit binary full adder
7485	4-bit magnitude comparator
7490	Decade ($\div 10$) counter
7493	4-bit binary ($\div 16$) ripple counter
74138	3-to-8 decoder
74147	10-to-4 priority encoder
74151	8-to-1 multiplexer
74153	Dual 4-to-1 MUX
74154	4-to-16 decoder
74157	Quad 2-to-1 MUX
74194	4-bit universal shift register